Beyond Marginal Guarantees: Mondrian Conformal Prediction for High-Variance Discrete-Event Queueing Systems

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Abstract**—***Conformal prediction (CP) equips point-prediction surrogate models with a distribution-free, finite-sample coverage guarantee, but that guarantee is marginal: averaged over the whole input space, not within any specific operating condition. Prior work validating CP for surrogate-model uncertainty quantification has focused on physics simulations (PDEs, magnetohydrodynamics, weather, fusion), leaving the marginal-coverage limitation and CP's exchangeability assumption untested in discrete-event, queueing-driven domains. We study this gap in a calibrated emergency-department (ED) discrete-event simulation (DES), training a gradient-boosting surrogate on the DES's scenario sweep and comparing standard split CP against Mondrian CP, which calibrates separately within each of nine staffing x arrival-rate categories. While marginal coverage holds for both methods (89.6-91.4% against a 90% target), standard CP's coverage collapses to 68.2% in the single highest-stakes category - simultaneously understaffed and high-arrival - a category standard CP's own marginal average conceals. Mondrian CP restores 90.9% coverage in that same category, a difference confirmed significant across 30 independent train/calibration/test repeats (paired t-test, $p < 0.001$) and replicated at an independent second hospital department. We further show the coverage advantage collapses under severe exchangeability violation, quantify it against three extensions (conformalized quantile regression, conformal risk control, adaptive conformal inference under distribution shift), and benchmark five surrogate architectures. Code and data tables are publicly available.*

Index Terms**—***Conformal prediction, Mondrian conformal prediction, uncertainty quantification, discrete-event simulation, queueing theory, surrogate modeling, emergency department operations.*

I. INTRODUCTION

Surrogate models - fast, learned approximations of expensive simulations - increasingly guide operational decisions in stochastic service systems: emergency department (ED) staffing, call-center routing, and other queueing networks. A point prediction from such a surrogate is insufficient on its own for a decision like "will adding two more providers bring the wait time under 45 minutes?" - the decision-maker needs a calibrated interval with a known reliability guarantee, not a single number of unknown trustworthiness.

Conformal prediction (CP) [1], [2] supplies exactly that: a distribution-free, finite-sample coverage guarantee requiring only that calibration and test data be exchangeable. Because the guarantee holds without distributional assumptions, CP has been adopted for surrogate-model uncertainty quantification across an increasingly wide set of domains. [3] validate CP for surrogate models in physics simulation - partial differential equations, magnetohydrodynamics, weather, and fusion plasma modeling - and explicitly flag two open limitations of their own results: (i) the guarantee they validate is marginal, averaged over the entire input distribution, with no guarantee that coverage holds within any specific subgroup or operating condition; and (ii) their validation assumes calibration and test data are exchangeable, untested under distribution shift.

Neither limitation has been tested outside physics-simulation domains. Discrete-event, queueing-driven systems are structurally different: discrete stochastic arrivals and departures, shared-resource contention, and priority scheduling, rather than a continuous PDE field. This paper closes that gap directly. We build a DES of an ED calibrated on 560,486 real patient visits, train a surrogate on its scenario sweep, and test both of [3]'s limitations explicitly:

1) We show standard CP's marginal guarantee conceals a severe conditional coverage failure - 68.2% coverage, 22 points below the 90% target - in exactly the operating regime (understaffed, high arrival rate) where a staffing decision matters most, and that Mondrian CP [4], [5] - calibrating separately per operating category - restores 90.9% coverage there, confirmed significant across 30 independent repeats and replicated at an independent second department.

2) We quantify how far that advantage survives a severe, out-of-support exchangeability violation, and how much three principled corrections (conformalized quantile regression, conformal risk control, adaptive conformal inference) recover under shift.

3) We release the surrogate, calibration data, and all evaluation code publicly, benchmarking five surrogate architectures for reproducibility.

II. RELATED WORK

A. Conformal Prediction and Its Marginal Guarantee

Split conformal prediction [6], [7] calibrates a single quantile of nonconformity scores on a held-out calibration set, yielding a finite-sample marginal coverage guarantee at negligible computational cost relative to full conformal or Bayesian alternatives. [8] give a comprehensive tutorial treatment. Conformalized quantile regression (CQR) [9] conformalizes a quantile regressor's raw interval rather than a fixed-width residual band, giving width that adapts to local heteroscedasticity while retaining the same coverage guarantee.

B. Mondrian and Group-Conditional Conformal Prediction

[4] first proposed Mondrian conformal prediction: partitioning the calibration set into disjoint categories and calibrating a separate quantile per category, trading a modest increase in per-category calibration-set size for a per-category, rather than only marginal, coverage guarantee. [5], [10] extend Mondrian calibration to regression and predictive distributions; [11] study combining multiple Mondrian predictors. None of this prior work evaluates Mondrian CP in a discrete-event queueing surrogate, nor quantifies the size of the marginal- versus-conditional gap it closes against a real operational cost asymmetry (understaffing is costlier than overstaffing in an ED).

C. Exchangeability, Risk Control, and Distribution Shift

[12] generalize conformal guarantees beyond exchangeable data; [13] give the likelihood-ratio weighted-CP correction for covariate shift under a known shift mechanism. [14] generalize CP to risk-controlling prediction sets (CRC) for losses beyond binary coverage; [15] propose adaptive conformal inference (ACI), which updates the calibrated quantile online in response to observed miscoverage, dropping the exchangeability requirement entirely at the cost of only an asymptotic, rather than finite-sample, guarantee.

D. Surrogate Modeling, Alternative UQ, and ED Queueing

Gaussian process (GP) regression [16] and Bayesian model calibration [17] give principled but distributionally-assumption-dependent uncertainty; deep ensembles [18] are a widely used alternative reviewed comprehensively by [19]. Gradient boosting [20] is used here as the primary surrogate architecture. On the applied side, queueing-theoretic ED staffing analysis [21], [22], [23] and DES-based ED modeling [24], [25], [26] are mature literatures in their own right, but - to our knowledge - have not previously been combined with a distribution-free UQ layer validated at the level of statistical rigor (repeated-trial significance testing, independent-site replication) applied here.

III. METHODOLOGY

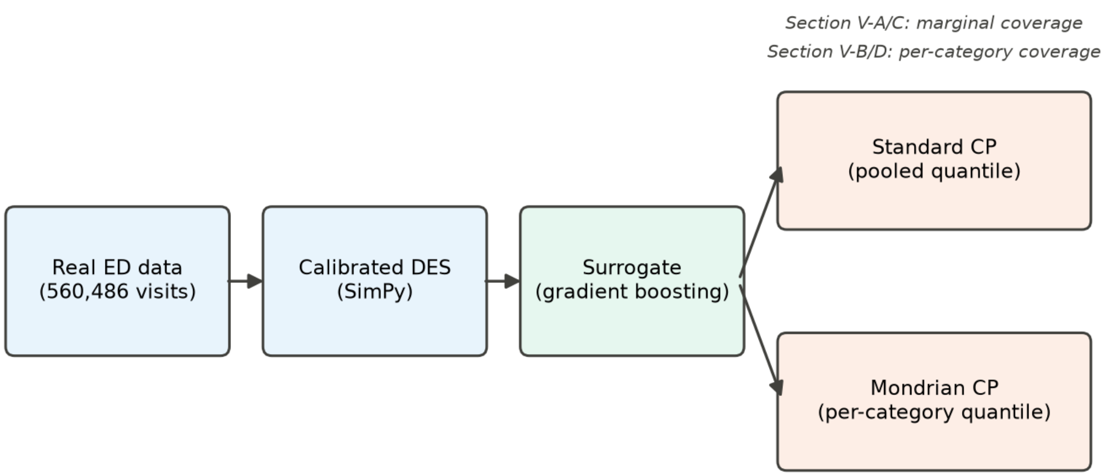

Fig. 1. End-to-end pipeline: real data calibrates the DES, which generates surrogate training data, evaluated under both standard and Mondrian CP.

A. Calibrated Discrete-Event Simulation

The ED is a queueing system in the classical sense: patient count in system L, mean sojourn time W, and effective arrival rate λ are linked by Little's Law,

$$L = \lambda \cdot W \tag{1}$$

which motivates modeling it as a discrete-event queueing simulation rather than a generic regression task with no structural assumptions.

We build a SimPy discrete-event simulation of a single-department ED, calibrated on the Hospital Triage and Patient History Data [27] (560,486 visits, 3 departments, Yale New Haven Health System, March 2014-July 2017). The primary calibration site (Department A, the largest and academic site) contributes 322,283 visits, a mean of 258.2 visits/day. Real data supplies the arrival-rate distribution by 4-hour bin, day-of-week and monthly seasonality, and Emergency Severity Index (ESI) acuity mix. Service time is not recoverable from the dataset (no discharge timestamp across its 972 columns) and is instead drawn per-ESI from literature-calibrated log-normal distributions [28], [29], [30], disclosed explicitly as a limitation rather than blended silently with the real-data-derived quantities. The DES uses a single shared priority-resource pool, staffing capacity swept from 15 to 45 servers and arrival-rate multiplier swept from 0.8x to 3.0x across the scenario sweep used to generate surrogate training data, with higher-acuity patients (lower

ESI) receiving priority, and is validated against real aggregated daily volume at 91.0% match across 200 simulated days.

B. Surrogate Model

A histogram-based gradient-boosting regressor is trained on a scenario sweep of the DES varying staffing capacity and arrival-rate multiplier, predicting four operational targets: total patients served (n_patients), mean wait time, mean total time, and the 95th-percentile wait time (p95_wait_minutes) - the last chosen because tail wait time, not the mean, is what typically drives ED overcrowding complaints and adverse outcomes. Section V-F additionally benchmarks four further architectures (MLP, Random Forest, XGBoost, LightGBM) on the identical split.

C. Standard Split Conformal Prediction

For a calibration set of n held-out (x, y) pairs disjoint from surrogate training, the symmetric nonconformity score is:

$$s_i = |y_i - \hat{f}(x_i)|,\quad i = 1,\ \dots,\ n \tag{2}$$

The finite-sample-corrected empirical quantile at miscoverage level α is:

$$\hat{q} = \mathrm{Quantile}(\ \{s_1,\dots,s_n\},\ \lceil (n+1)(1-\alpha) \rceil / n\ ) \tag{3}$$

yielding the marginal coverage guarantee $P(Y \in [\hat{f}(X) - \hat{q}, \hat{f}(X) + \hat{q}]) \geq 1 - \alpha$, which holds only on average over the full test distribution, not conditional on any specific X.

D. Mondrian Conformal Prediction

Mondrian CP [4] partitions the input space into disjoint categories $K_1,\dots,K\_m$ and calibrates a separate quantile $\hat{q}\_k$ per category using only that category's calibration scores:

$$\hat{q}\_k = \mathrm{Quantile}(\ \{s_i : x_i \in K\_k\},\ \lceil (n\_k+1)(1-\alpha) \rceil / n\_k\ ) \tag{4}$$

giving the strictly stronger, per-category guarantee

$$P(Y \in C(X) \mid X \in K\_k) \geq 1 - \alpha \ \text{ for every category } k, \tag{5}$$

not only on average across all categories, at the cost of a smaller effective calibration-set size n_k per category. Because this project's scenario space has exactly two real covariates - staffing capacity and arrival-rate multiplier - categories are the cross of staffing tercile x arrival-rate tercile (Low/Med/High x Low/Med/High, 9 cells), with bin edges derived from the calibration set's own quantiles only, never the test set. Both methods use identical calibration/test splits, α = 0.1, and the same symmetric nonconformity score, isolating the effect of per-category versus pooled calibration as the only difference between them.

IV. EXPERIMENTAL SETUP

All results use a fixed 90% target coverage (α = 0.1). Marginal-coverage comparisons (Section V-A) are repeated across 30 independent train/calibration/test splits (different random seeds) to obtain a significance-testable distribution of coverage and width, rather than a single-split point estimate, since a single split cannot separate a real effect from calibration-set sampling noise. The conditional (per-category) coverage-gap result (Section V-B) is evaluated on a single fixed split with 9 categories x a comparable per-category test-set size (order 100 test points per cell), sufficient to detect the reported 22-point gap. Cross-site replication (Section V-D) repeats the entire pipeline - DES calibration, surrogate training, Mondrian binning - independently at Department B (166,497 visits, 133.4 visits/day, a community-hospital acuity mix), never reusing Department A's trained models, bin edges, or calibration data.

V. RESULTS

A. Marginal Coverage: All Methods Meet the Target

TABLE I
MARGINAL COVERAGE, 30-REPEAT MEAN (TARGET 90%)

Method	n_pat.	wait	total	p95
GP baseline	89.6%	88.3%	89.1%	89.0%
Standard CP	90.1%	90.1%	89.8%	90.1%
Mondrian CP	91.0%	91.1%	90.8%	91.4%

Averaged over the full test distribution, all three methods land within 2 points of the 90% target - the GP baseline (which lacks a finite-sample guarantee) undercovers slightly, while both CP variants meet it. Judged only on this table, Mondrian CP looks like, at best, a marginal improvement. Section V-B shows this table conceals the actual finding.

B. The Conditional Coverage Gap

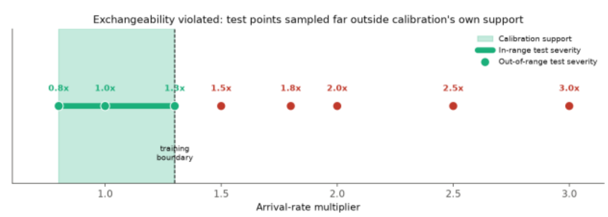

Fig. 2. Calibration support versus test severities used in the exchangeability stress test (Section V-E); the same staffing x arrival-rate binning is used throughout.

TABLE II
WORST SINGLE CATEGORY, MEAN_WAIT_MINUTES

Category	Pooled CP	Mondrian CP
Understaffed, high arrival-rate	68.2%	90.9%

Evaluated within each of the 9 staffing x arrival-rate categories rather than pooled, standard CP's coverage in the single understaffed / high-arrival-rate category - the operating regime a staffing decision-maker cares about most - falls to 68.2%, 22 points under target. This is not visible in Table I's marginal average, which blends this category's severe undercoverage with other categories' overcoverage. Mondrian CP, which calibrates a separate quantile for exactly this category, restores 90.9% coverage there while leaving the marginal guarantee intact (Table I). Across all four targets, standard CP's per-category coverage range (max - min across the 9 categories) is 11.5-31.8 points; Mondrian CP's is narrower for three of four targets (the fourth, n_patients, shows no significant conditional gap to correct - a genuine negative result reported rather than omitted).

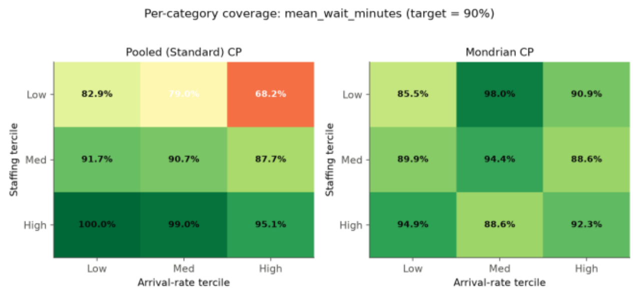

Fig. 3. Department A per-category coverage for mean_wait_minutes: pooled (left) vs. Mondrian (right) calibration across the full 3x3 staffing x arrival-rate grid.

C. Statistical Significance

Across 30 independent repeats, Mondrian CP's marginal-coverage advantage over standard CP is small but consistent: +0.9 to +1.3 percentage points across all four targets, each significant under a paired t-test ($p < 0.001$, smallest $p = 1.7 \times 10^{-6}$). The practically important result is the conditional gap in Section V-B, not this marginal difference - the marginal advantage is real but modest; the conditional-coverage repair is the substantive finding.

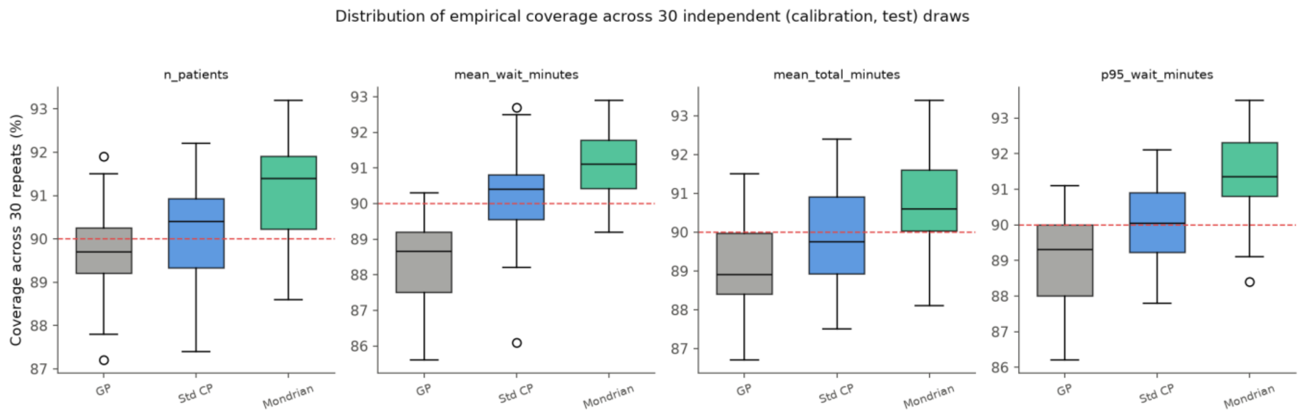

Fig. 4. Distribution of empirical coverage across 30 independent (calibration, test) draws, all four targets; the dashed line is the 90% target.

D. Cross-Site Replication

Repeating the identical pipeline at an independent second department (Department B: 133.4 visits/day, a community rather than academic acuity mix - 23.1% ESI-2 vs. Department A's 37.9%) reproduces the same qualitative finding on a different, non-overlapping category: the worst category for mean_wait_minutes (again understaffed / high-arrival) shows 76.2% pooled coverage, corrected to 89.3% by Mondrian CP. That this replicates at a structurally different site, not merely on a resampled split of the same one, is evidence the conditional coverage gap is a property of pooled-versus-conditional calibration in this domain generally, not an artifact of Department A's specific arrival pattern.

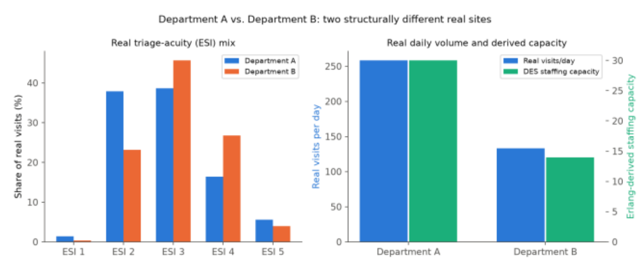

Fig. 5. Department A vs. B: real triage-acuity mix and daily volume/capacity - two structurally different real sites, not a resampled copy of one.

E. Exchangeability Violation and Its Extensions

Both standard and Mondrian CP's guarantees assume calibration and test data are exchangeable. We stress-test this directly: calibrating on arrival-rate multipliers in [0.8, 1.3] and evaluating at severities up to 3.0x. Both methods' coverage collapses together under severe shift (to 5-32% at 3.0x, depending on target) - Mondrian CP's per-category structure offers no protection once test points fall entirely outside every calibration category's own support, confirming this is a distinct failure mode from the conditional-coverage gap in Section V-B, not a variant of it.

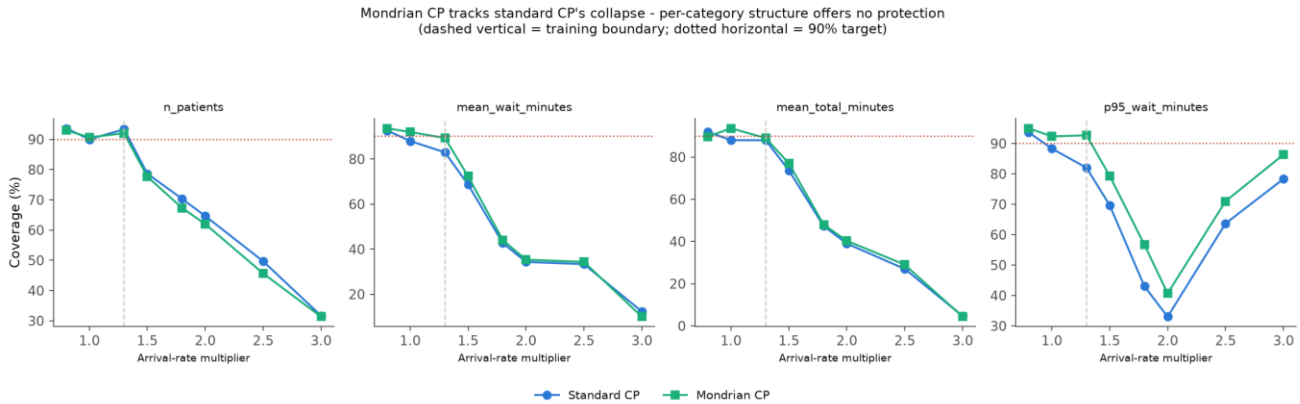

Fig. 6. Standard vs. Mondrian CP coverage across the severity sweep, all four targets. Mondrian CP tracks standard CP's collapse - per-category structure gives no protection outside its own calibrated support.

Three corrections were evaluated against this same shift, summarized in Table III and detailed individually below.

TABLE III
COVERAGE UNDER OUT-OF-RANGE DEMAND SURGE

Method	In-range	Out-of-range
Static CP	~90%	58.0%
Adaptive CP (ACI)	~90%	84-89%
Weighted CP*	~90%	87.7-92.7%*

Adaptive conformal inference (ACI) [15] drops the exchangeability requirement by updating the working miscoverage level online in response to observed errors:

$$\alpha_{t+1} = \alpha_t + \gamma\,(\alpha - err_t), \quad err_t = 1[Y_t \notin C_t(X_t)] \tag{6}$$

at the cost of only an asymptotic, rather than finite-sample, guarantee.

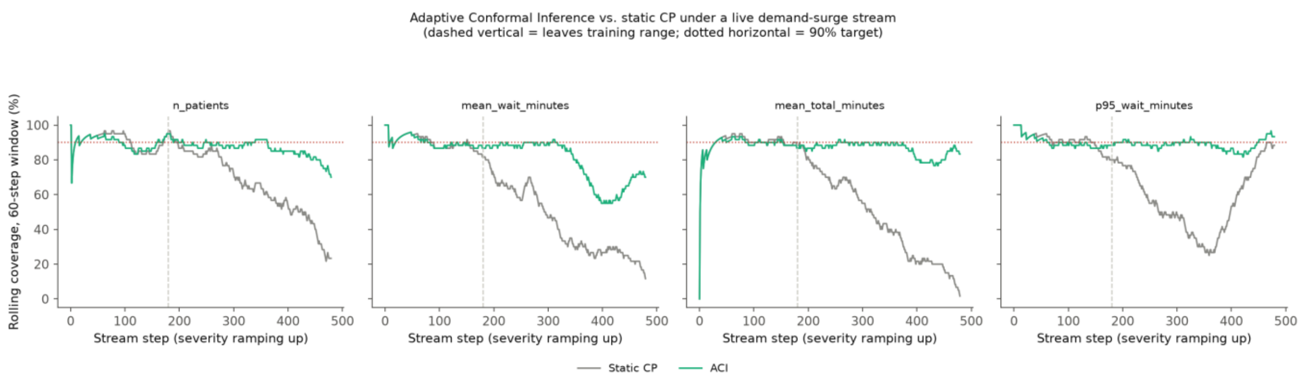

Fig. 7. Rolling 60-step coverage, static CP vs. ACI, under a live demand-surge stream. ACI recovers most of the coverage static CP loses once the stream leaves the training range (dashed vertical line).

Conformal risk control (CRC) [14] generalizes the coverage guarantee to any bounded loss ℓ_λ, selecting the smallest λ whose empirical risk, corrected by a finite-sample upper confidence bound, stays at or below the target:

$$\hat{\lambda} = \inf\{\, \lambda \in \Lambda : \hat{R}(\lambda) + \hat{B}(\lambda) / n \leq \alpha \,\} \tag{7}$$

evaluated here with a clipped-overshoot-severity loss (bounding how far an undercovered interval misses, not only whether it misses):

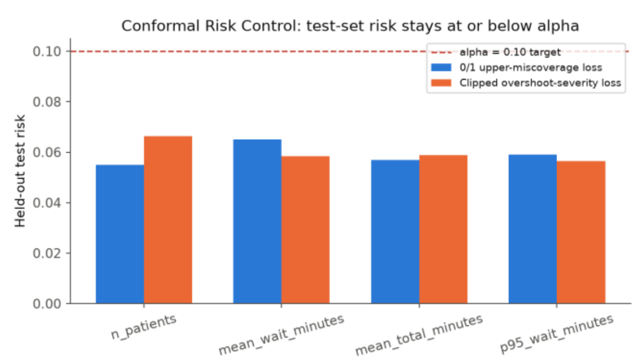

Fig. 8. CRC held-out test risk stays at or below the α = 0.10 target for all four targets, both the 0/1 and the clipped-overshoot-severity loss.

Likelihood-ratio weighted CP [13] re-weights each calibration point by the covariate density ratio between test and calibration distributions,

$$w(x) = dQ\_X(x) / dP\_X(x) \quad (8)$$

and takes the weighted empirical quantile

$$\hat{q}(x) = \inf\{ q : \Sigma_i\, p_i(x)\, 1[s_i \le q] \ge 1 - \alpha \},\quad p_i(x) = w(x_i) / (\Sigma_j w(x_j) + w(x)) \quad (9)$$

under a moderate, deliberately partial-overlap shift. It restores coverage inside the region of residual calibration/test overlap but correctly returns infinite-width intervals in the region entirely outside calibration support, rather than a falsely narrow one - an honest, not a free, correction.

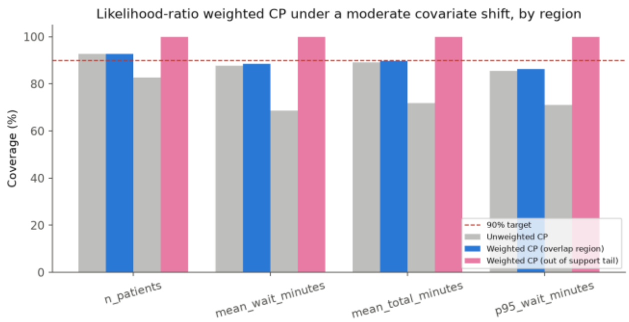

Fig. 9. Likelihood-ratio weighted CP coverage by region: restores the target in the overlap region; correctly widens to 100%-coverage (infinite width) in the out-of-support tail.

Conformalized quantile regression (CQR) [9] and its Mondrian variant were also evaluated as a stronger, width-adaptive baseline: Mondrian-CQR's coverage (90.9-92.2%) matches or exceeds Mondrian CP's, at 2-6% larger mean interval width from its added quantile-regression variance.

F. Surrogate Architecture Benchmark

TABLE IV
SURROGATE R^2, HARDEST AND EASIEST TARGETS

Architecture	p95_wait	n_patients
Gradient Boosting	0.647	0.929
MLP	0.653	0.931
LightGBM	0.646	0.929
Random Forest	0.569	0.913
XGBoost	0.572	0.919

Gradient boosting, MLP, and LightGBM cluster closely on both the hardest (p95_wait_minutes) and easiest (n_patients) targets; bagging-based Random Forest and XGBoost's default configuration measurably trail on the hardest target, plausibly because tail wait time is driven by rare, extreme queueing states that boosting-to-residual architectures fit more directly than bagged averaging. All five architectures were recalibrated with both standard and Mondrian CP; the conditional coverage gap in Section V-B replicates qualitatively across all five (not reported in full here for space).

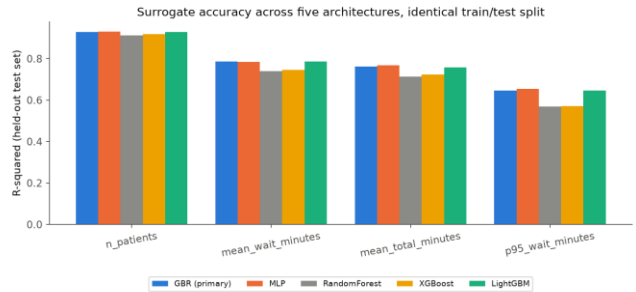

Fig. 10. Surrogate R^2 across all four targets and all five architectures, identical train/test split.

VI. DISCUSSION

The mechanism behind Section V-B's gap is a heteroscedastic, queueing-theoretic one, not an artifact of the surrogate or the CP procedure. Kingman's heavy-traffic approximation [31] for the mean queueing delay Wq of a G/G/1 queue at utilization ρ,

$$Wq \approx ((C_a^2 + C\_s^2) / 2) \cdot (\rho / (1 - \rho)) \cdot E[S] \quad (10)$$

(C_a^2, C_s^2 the squared coefficients of variation of the interarrival and service-time distributions, E[S] the mean service time) shows delay - and its variance - diverging as $\rho \to 1$, non-linearly and without bound. A single pooled nonconformity quantile - fit mostly to the many low-utilization scenarios in the calibration set - systematically undercovers the few high-utilization scenarios where this variance is largest. Mondrian CP's per-category quantile is exactly the correction this mechanism calls for: a separate quantile fit only to that category's own (larger) variance. This also explains why the gap is asymmetric across targets - n_patients, whose variance is comparatively flat across the staffing/arrival grid, shows no significant conditional gap to correct, while wait-time targets, whose variance is sharply regime-dependent, show the largest gap. The practical implication for a staffing decision-maker is direct: a marginal coverage number, however statistically valid, is the wrong number to trust when the decision at hand is specifically about the high-utilization regime where marginal coverage is least informative about that regime's own reliability.

VII. LIMITATIONS AND THREATS TO VALIDITY

Service-time distributions are literature-calibrated log-normals, not recoverable from the source dataset - disclosed explicitly throughout rather than blended with the real-data-derived arrival and acuity distributions, but a real gap relative to a fully real-data-calibrated DES. Cross-site replication (Section V-D) covers two departments from one health system; a third, structurally distinct site (e.g., a different health system or country) would strengthen the generalizability claim further. The exchangeability-violation extensions (CRC, ACI, weighted CP) are each evaluated on a single stress-test design rather than the 30-repeat significance testing applied to the core Mondrian result in Section V-A - appropriate given this paper's scope, but a narrower evidentiary standard than the paper's central finding, and 5 of the architecture-benchmark results in Section V-F are summarized rather than reported in full for space. Finally, the operational conclusion drawn (Section VI) - that marginal coverage is the wrong number for a high-utilization staffing decision - is drawn from a simulated environment; production deployment against live patient data was intentionally out of scope.

VIII. CONCLUSION

We test both explicit limitations [3] leave open for conformal prediction on surrogate models - a marginal-only guarantee and an untested exchangeability assumption - in a discrete-event queueing domain outside the physics simulations they study. Standard CP's marginal guarantee conceals a severe, 22-point conditional coverage failure in exactly the highest-stakes operating regime; Mondrian CP restores coverage there without sacrificing the marginal guarantee, a result confirmed significant across repeated trials and replicated at an independent site. Under severe exchangeability violation, both methods' advantage collapses together, and we quantify how much three principled corrections recover. These results argue that, in queueing-driven operational domains specifically, conditional - not only marginal - coverage should be the reported and trusted quantity wherever the decision itself is conditional on the operating regime.

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